

SLDP Protocol Specification

Stateless Layer-2 Discovery Protocol for Cross-Family IPv4/IPv6 Bootstrapping

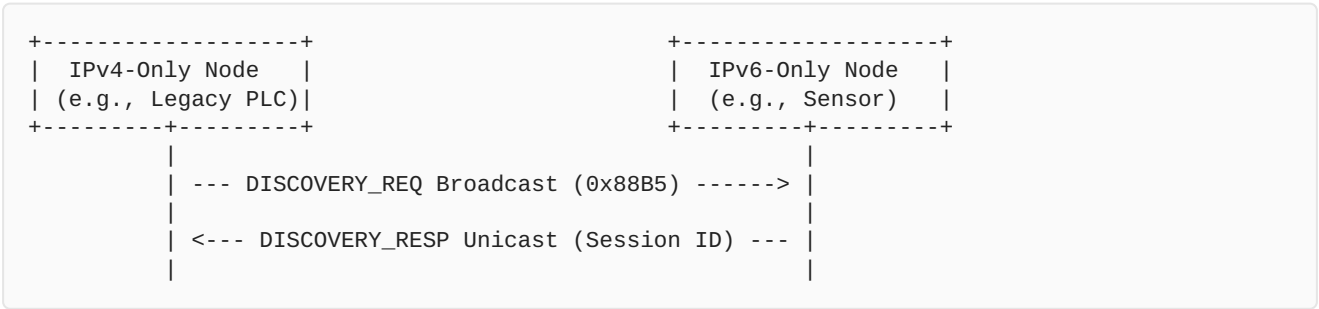
SLDP is a lightweight, stateless Layer-2 bootstrapping protocol designed to automate the discovery phase between IPv4-only and IPv6-only devices sharing a common broadcast domain. It operates directly over Ethernet frames—requiring zero IP stack initialization, active sockets, DNS, or DHCP infrastructure.

Note: SLDP is **not** a replacement for ARP, NDP, mDNS, or LLDP in general-purpose networks, nor is it a runtime protocol for time-sensitive production data planes. It is purpose-built strictly for the initial commissioning and bootstrapping phase.

Problem Statement & Overview

Modern network environments frequently host a hybrid mix of legacy IPv4-only hardware (e.g., PLCs, industrial relays) and modern IPv6-only devices (e.g., IoT sensors, robotic cameras). Historically, integrating these address families requires manual recording of MAC addresses, hand-coded proxy mappings, and static NAT64 table configuration—a workflow prone to typographical errors and operational downtime.

SLDP acts as an **on-demand commissioning protocol**. It allows cross-family devices to perform a stateless Layer-2 handshake to declare their MAC-to-IP capabilities prior to IP stack configuration. Once discovery completes and static mapping tables are built, SLDP stops transmitting.



Key Features & Protocol Architecture

- **Pure Layer-2 Execution:** Uses dedicated EtherType 0x88B5, bypassing the upper IP/UDP/TCP network stack.
- **Zero Infrastructure Dependencies:** Requires no active DHCP servers, DNS/mDNS resolution, or routing layers.
- **Stateless 2-Way Handshake:** Uses an atomic DISCOVERY_REQ broadcast and DISCOVERY_RESP unicast exchange.
- **Bootstrap vs. Runtime:** Runs on demand during commissioning or maintenance, then remains silent so it doesn't introduce overhead into production traffic.

How SLDP Compares to Existing Protocols

Protocol	Layer	Primary Function	Active Cross-Family Discovery?	Zero IP Stack Required?	Runtime vs Bootstrap
ARP / NDP	Layer 2/3	Single-family address resolution	No	No	Runtime (Continuous)
mDNS / DNS-SD	Application	Service & name resolution	No	No	Runtime
LLDP / CDP	Layer 2	Capability advertisement	No (Passive ads only)	Yes	Runtime (Periodic)
SLDP	Layer 2	Cross-family discovery	Yes	Yes	Bootstrap (On-Demand)

Protocol Frame Specification

SLDP frames pad to a standard 64-byte minimum Ethernet payload size.

Field	Length	Description
EtherType	2 Bytes	Defined as 0x88B5.
Version	1 Byte	Protocol version (Default: 0x01).
Message Type	1 Byte	0x01 (DISCOVERY_REQ), 0x02 (DISCOVERY_RESP).
Flags	2 Bytes	Control flags.
Session ID	4 Bytes	Transaction identification number.
Source MAC	6 Bytes	Hardware Ethernet address.
Source IP	16 Bytes	Native IPv6 or IPv4-mapped IPv6 address.

Building & Usage

Prerequisites

- Linux-based OS with gcc installed.
- root or CAP_NET_RAW privileges (required to open AF_PACKET raw sockets).

Compilation

```
gcc -O2 sldp.c -o sldp
```

Running the C Engine

Node 1: Listening Mode (e.g., IPv6-only Sensor)

```
sudo ./sldp eth0 fe80::10 listen
```

Node 2: Probing Mode (e.g., IPv4-only Gateway)

```
sudo ./sldp eth0 192.168.1.10 probe
```

Target Deployment Scenarios

- **Brownfield Industrial Automation:** Interfacing legacy IPv4 PLCs with newer IPv6 field sensors without manual static mapping.
- **Tactical & Disaster Recovery Networks:** Establishing rapid device visibility in field environments devoid of DNS/DHCP infrastructure.
- **Autonomous Swarms:** Ad-hoc mesh join/leave actions for drones and uncrewed ground vehicles operating in GPS-denied environments.
- **Data Center Bare-Metal Provisioning:** Out-of-band management bootstrapping during early PXE boot stages.